ICT30016 IoT Track – Workshop

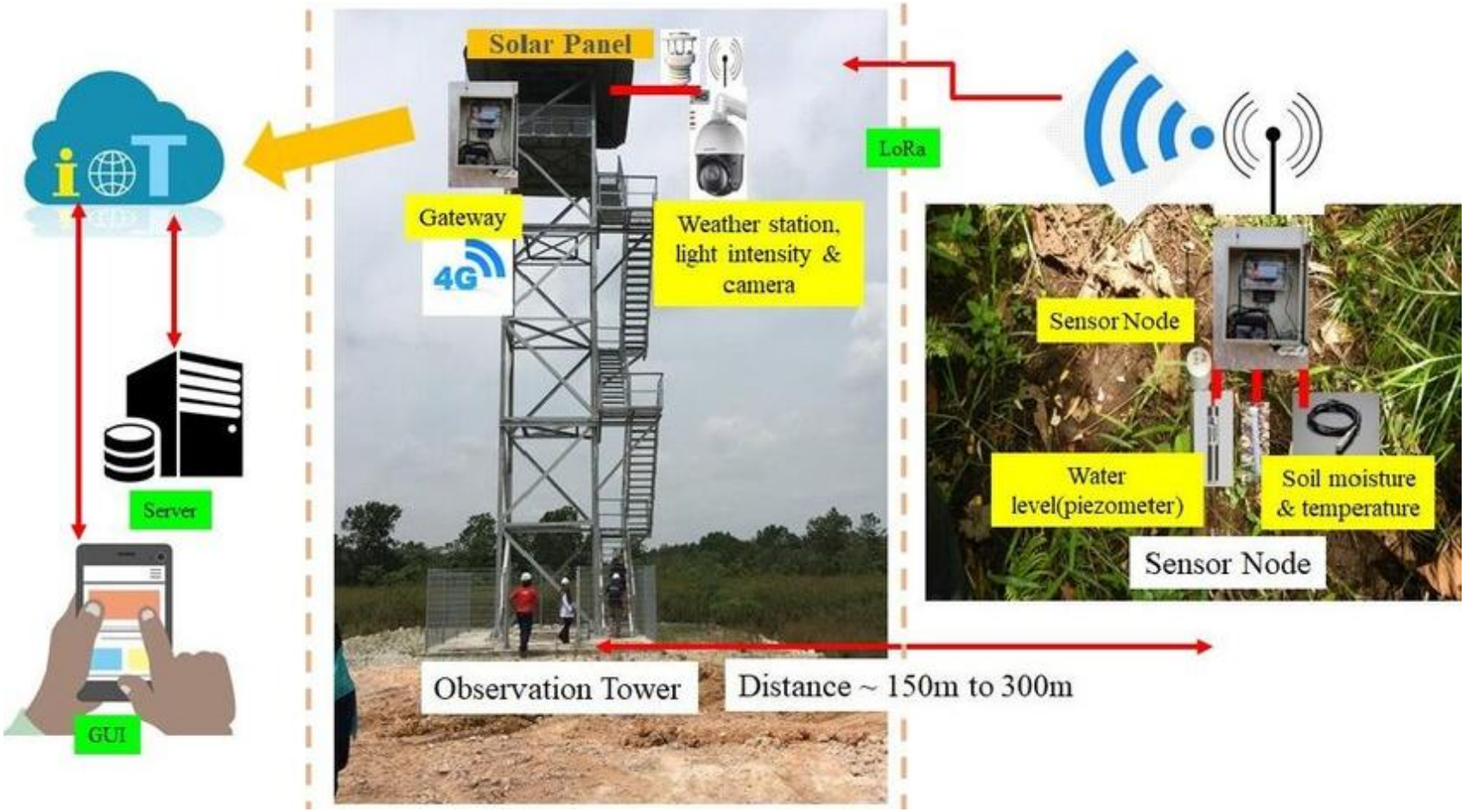
Putting it all Together

Prepared by Dr. Mark Tee Kit Tsun (SUTS)

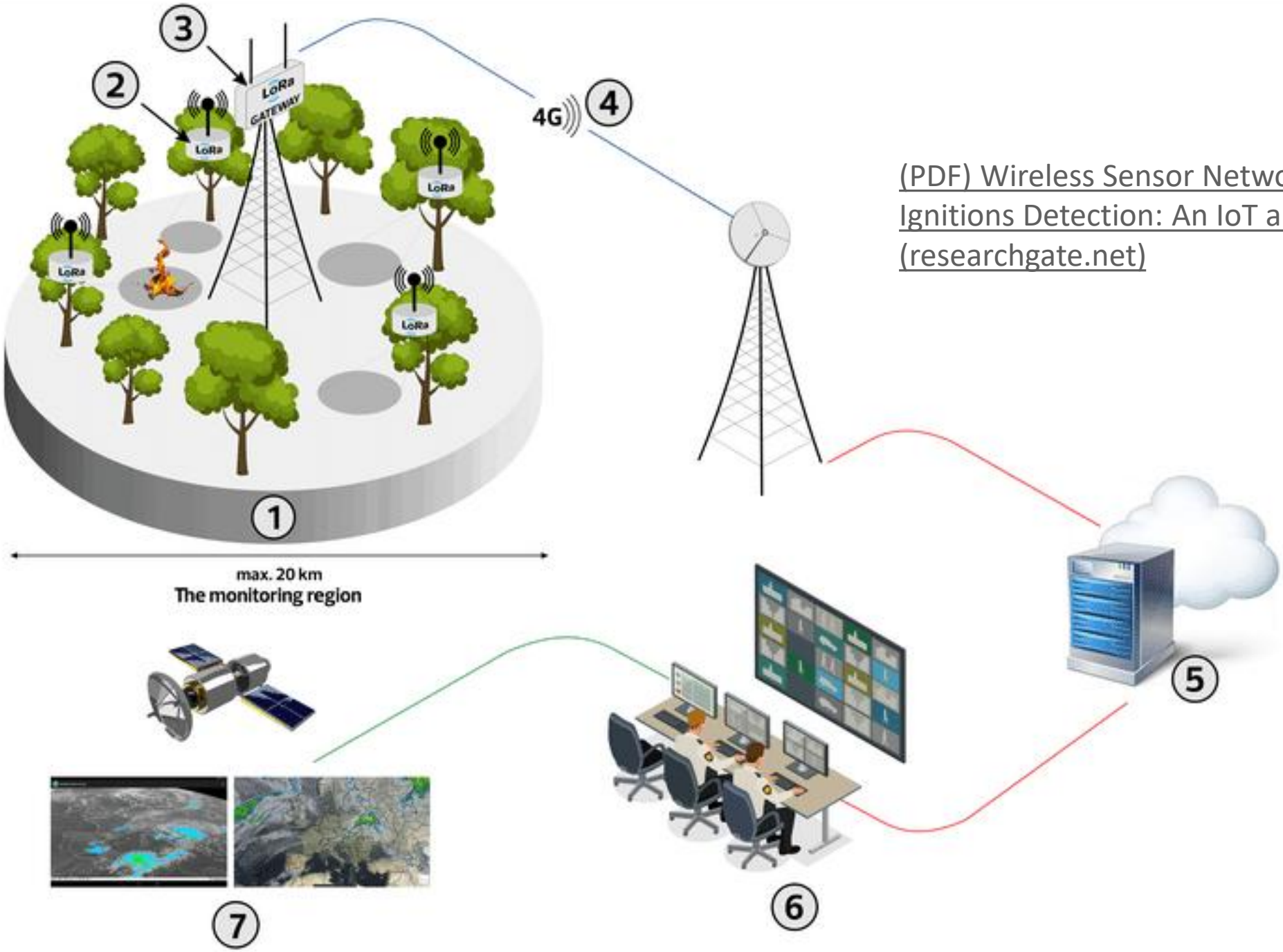
Overview of an IoT System

There are 3 characteristics of every IoT sub-system:

A. Each Node is fully functional (Computing + Sensors + Actuators + Communications + Power Supply)
B. Consists of **multiple** Nodes connected using some form of Networking
C. Connects to the Internet for Cloud Services (Frontend Dashboard, Database, Analytics, etc.)

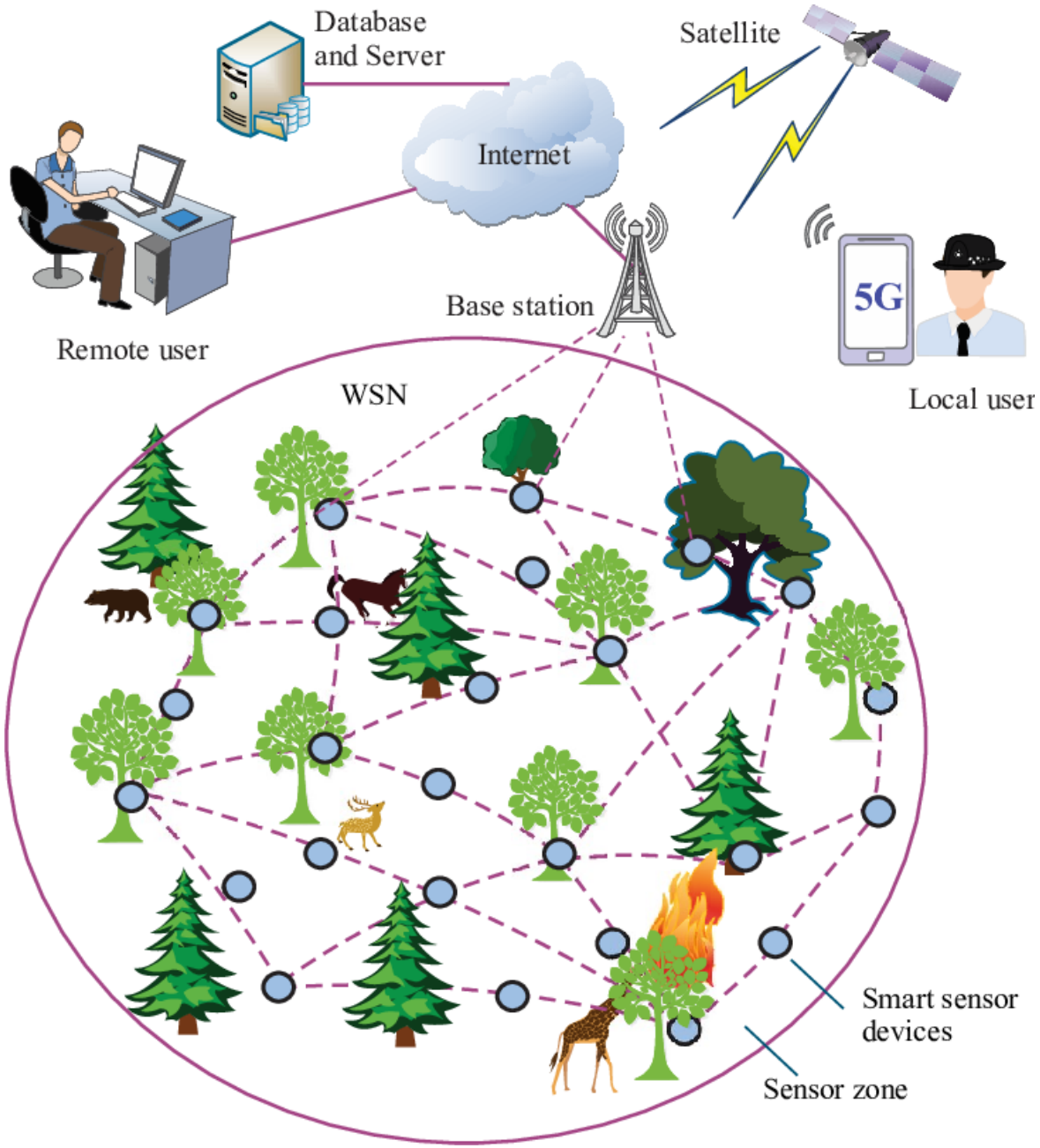


(PDF) Machine Learning for Peatland Ground Water Level (GWL) Prediction via IoT System (researchgate.net)



(PDF) Wireless Sensor Network for Ignitions Detection: An IoT approach (researchgate.net)

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Database
and Server
Satellite
Internet
Base station
5G
Remote user
Local user
WSN
Smart sensor
devices
Sensor zone



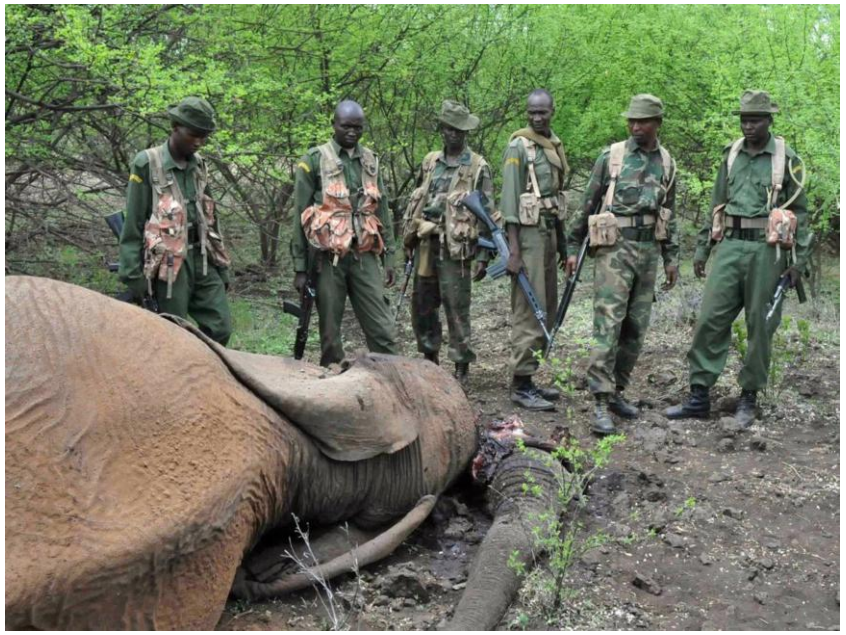






Case Study: Using IoT to Fight Poachers?









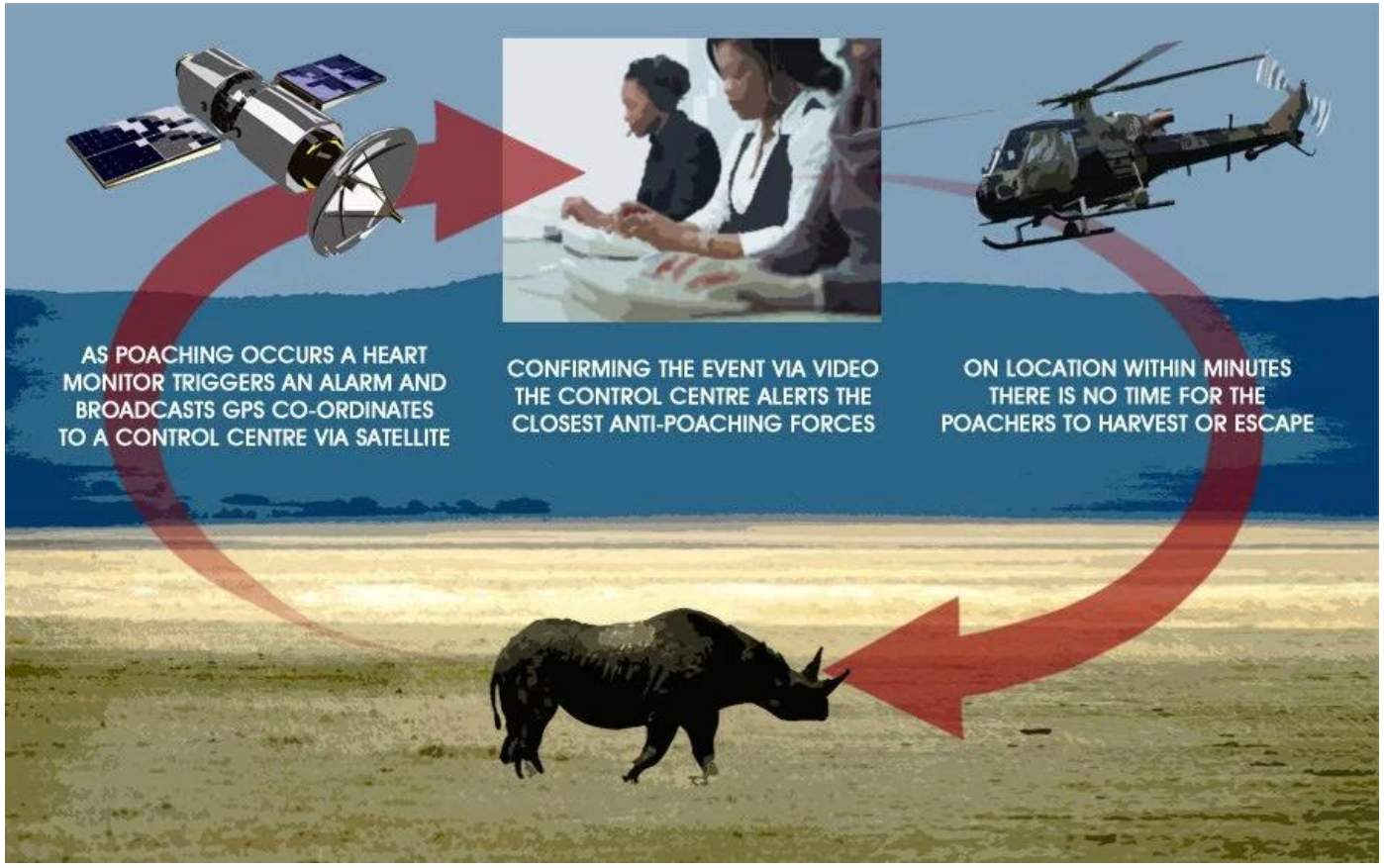




It is an Ongoing IoT Research Effort

Check out these articles:

- Catching poachers was almost impossible until this Dutch company invented a new type of camera
- How the Internet of Things Protects Wild Animals from Poachers (nutanix.com)
- How IOT can help stop animal Poaching? | LinkedIn





Iot Based Anti-Poaching Alarm System For Trees In Forest Using Wireless Sensor Networks (youtube.com)

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HOW IOT CAN SOLVE OUR ECOLOGICAL ISSUES - BIODIVERSITY

HEART RATE MONITOR

DRONES

ANIMAL TRACKING COMBINED WITH GPS TECHNOLOGY

WEATHER MONITORING SENSORS

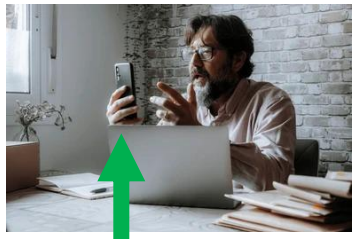
CONNECTED HABITATS

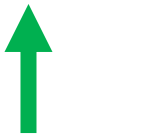
ACOUSTIC MONITORING

VISUAL & THERMAL IMAGING TECHNOLOGIES

For now, assume there are 2 kinds of Nodes:









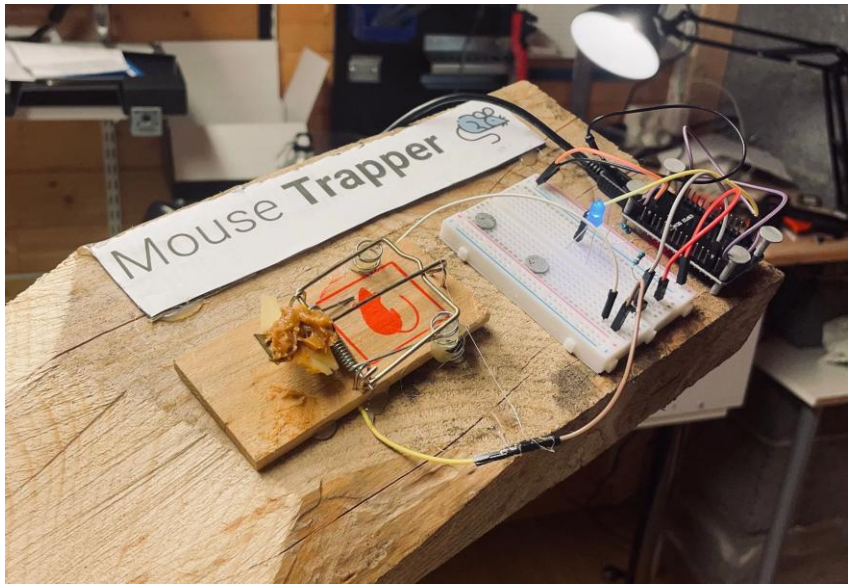
“Computing” Node

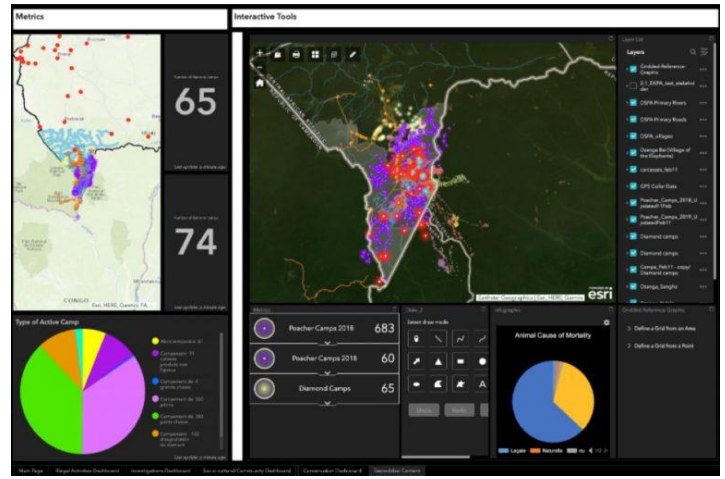


USB



“Automation” Node





WIFI

WIFI



Gateway



Internet Service Provider (ISP)



What is the Difference?

"**Computing**" Node

- The heart is a Microprocessor
  - Capable of running an OS
  - Can program and be programmed
  - Can run multiple software

- Can interface with many complex peripherals
  - Using USB – Cams, Printers, other Microcontrollers
  - Using PCI Express – Video Capture, Audio Cards, SSD

- Examples:









"**Automation**" Node

- The heart is a Microcontroller
  - Runs on a "firmware" image
  - Can be reprogrammed by another Computer
  - Only runs one program

- Can interface with Physical Layer electronics hardware
  - Buttons, Switches, Thermometers, and other Sensors
  - Trigger Motors, Lights, and other Actuators

- Examples:







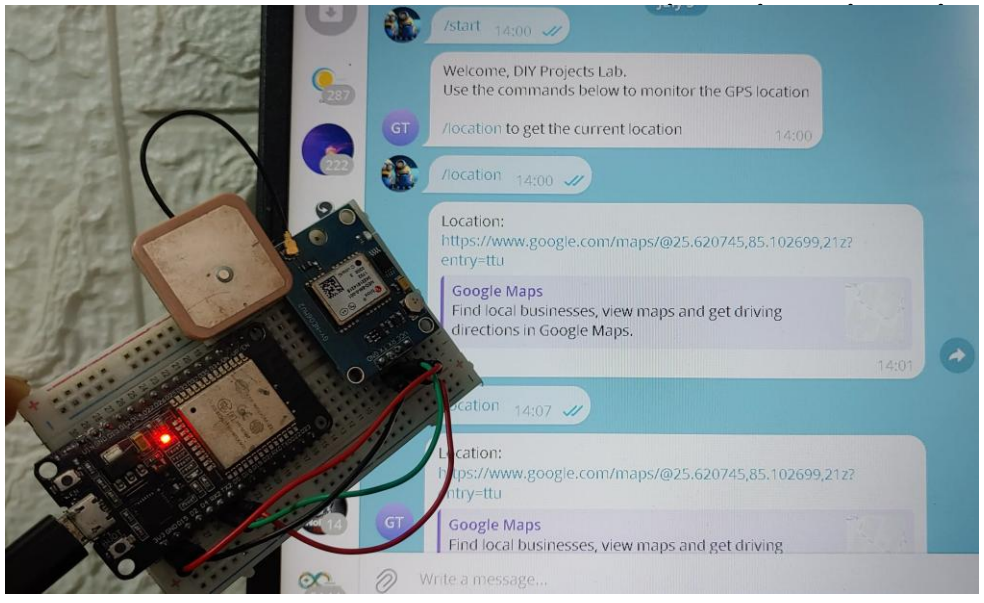


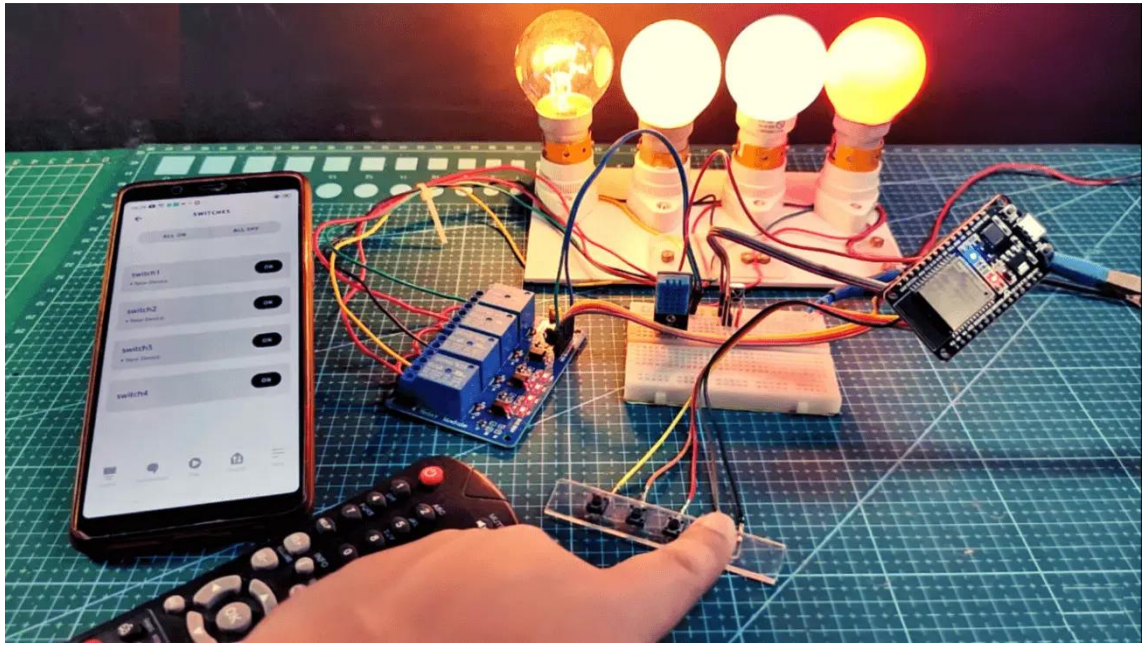
What Can You Do with the ESP32? Make an Automation Node!

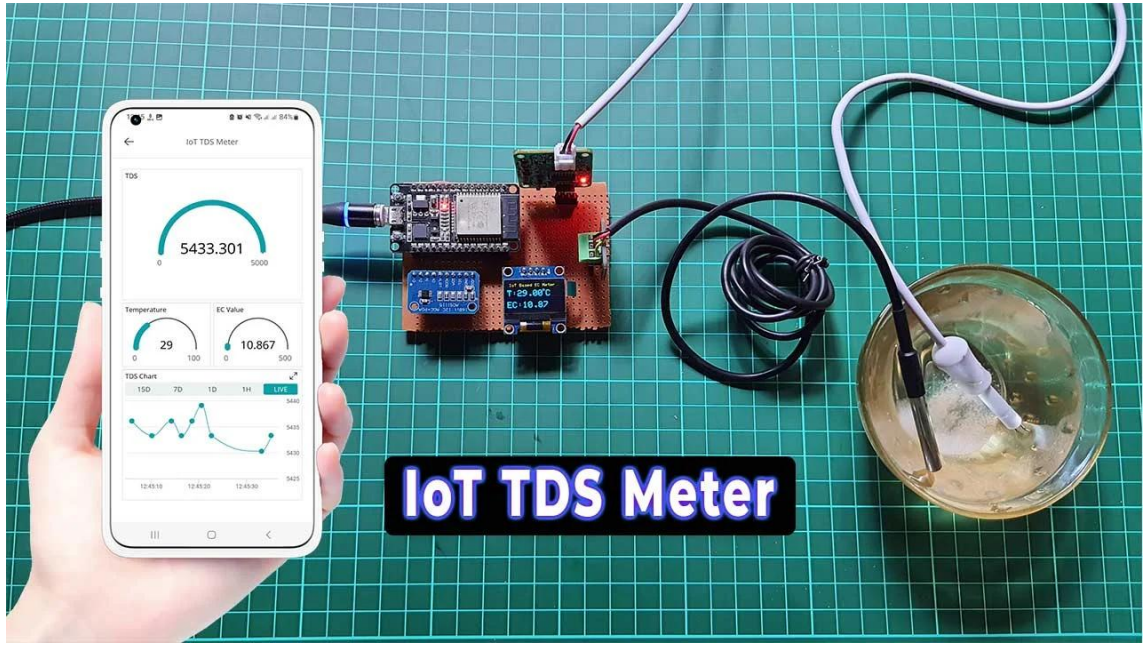
- NodeMCU ESP32 board is a Microcontroller (NOT a Microprocessor)!

- Good for creating automated systems like smart door locks, washing machines, remote light control, emergency response, etc.

- Its top features are:
  - The Ability to connect to WIFI & Bluetooth
  - 32-bit CPU to run more operations-intensive work
  - Arduino compatible
  - Cost-effective (Each board costs RM29)





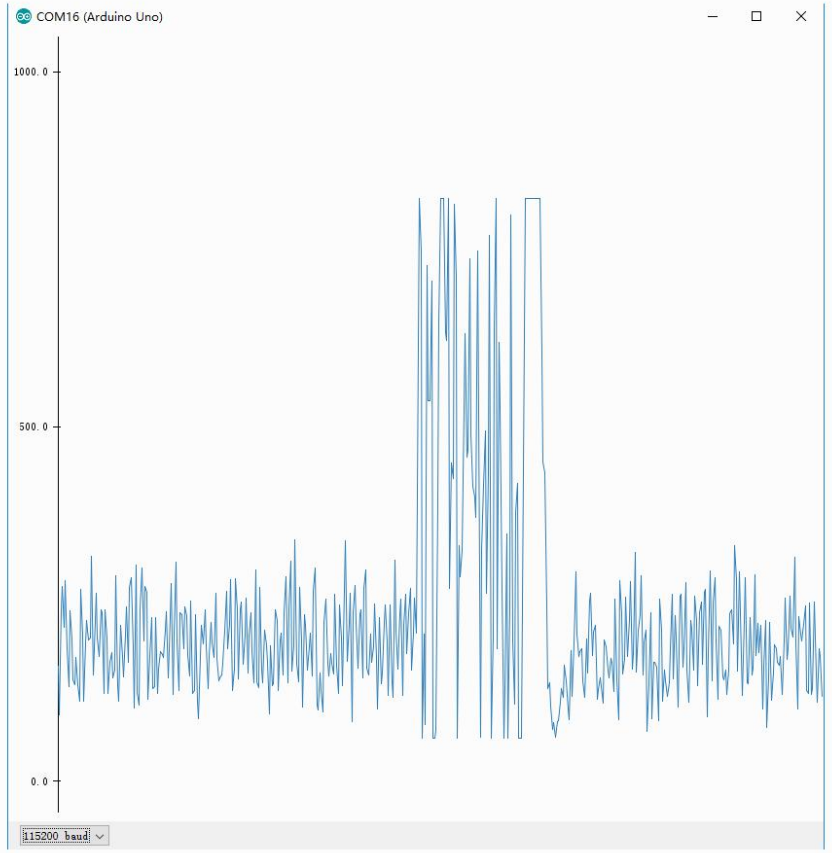


What Can You Do with the ESP32?



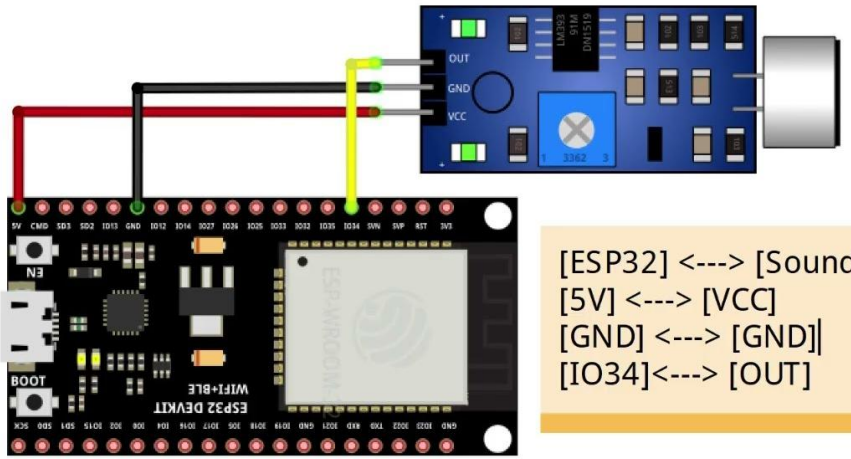
Understand what it can (and cannot) do. Examples:

- Using a Sound Sensor => It can measure Noise Levels (but cannot record audio clips)



ESP8266 NodeMCU – KY-038 Sound Sensor (geekering.com)



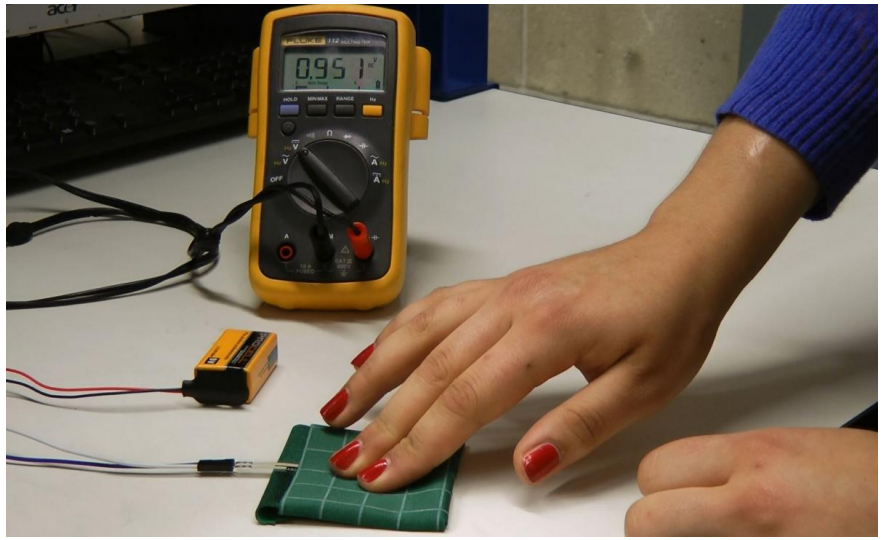


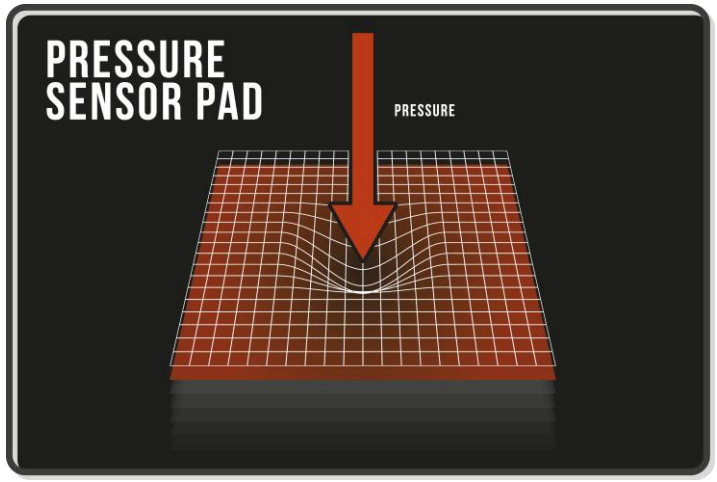
[ESP32] <---> [Sound sensor]
[5V] <---> [VCC]
[GND] <---> [GND]
[IO34]<---> [OUT]

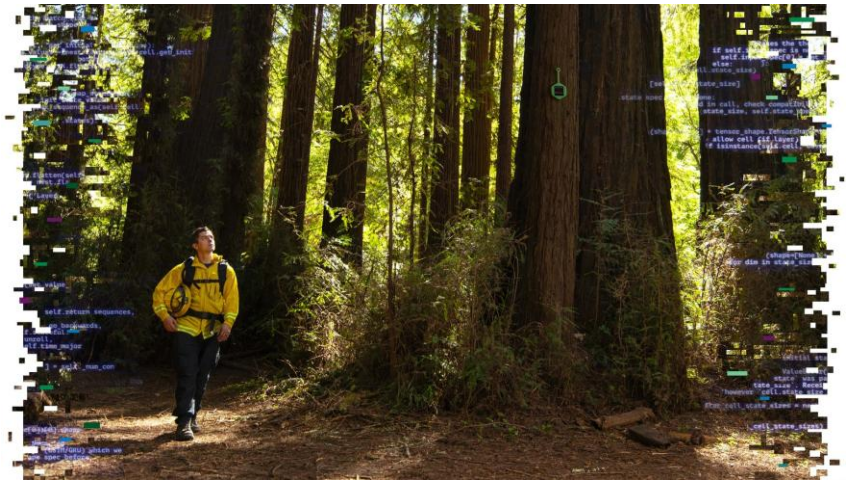
What Can You Do with the ESP32?

ESP32 - Button | ESP32 Tutorial (esp32io.com)

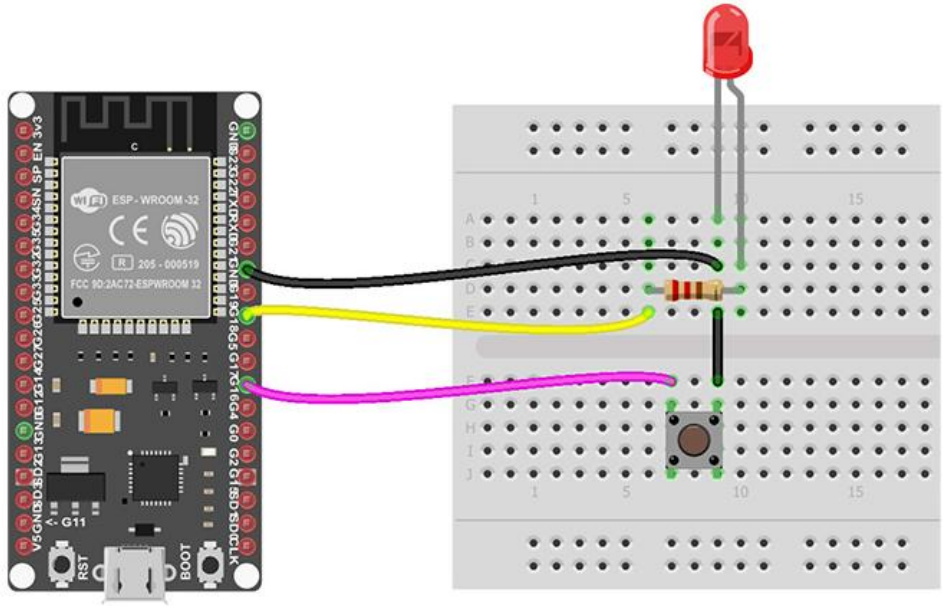
- Understand what it can (and cannot) do. Examples:
  - Using a Button => It can represent a pressure sensor to see if someone steps on it









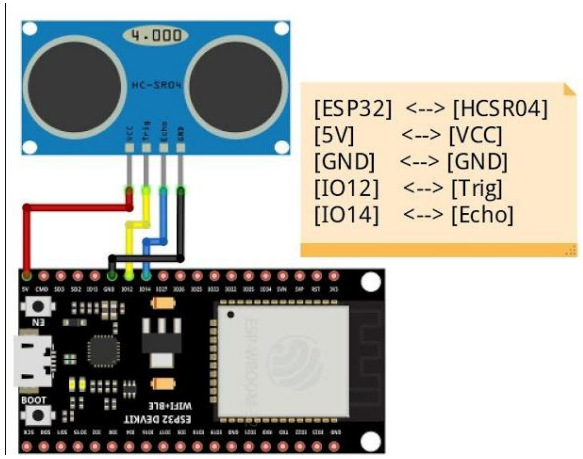


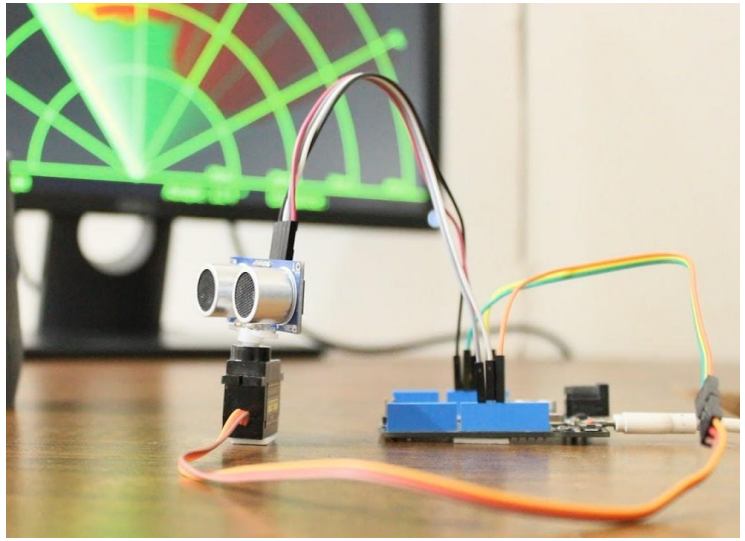
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What Can You Do with the ESP32?

- Understand what it can (and cannot) do. Examples:
  - Using an Ultrasonic Sensor => Detect if something is coming near it

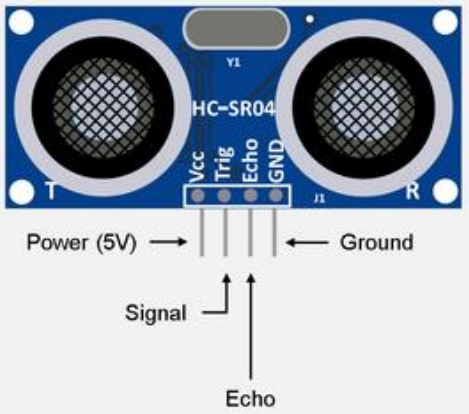
HCSR04 ultrasonic sensor - Arduino Reference

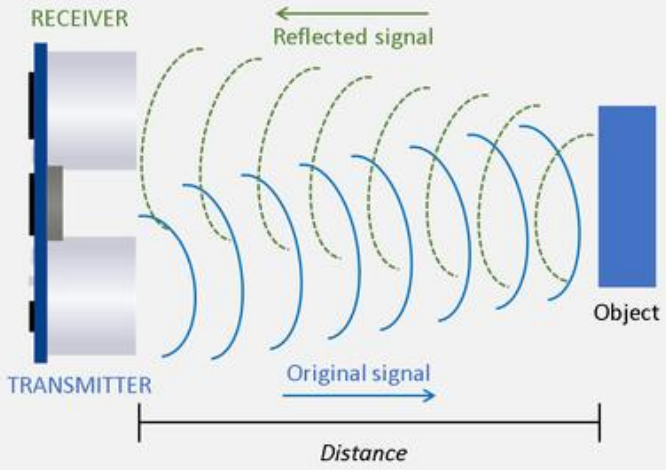








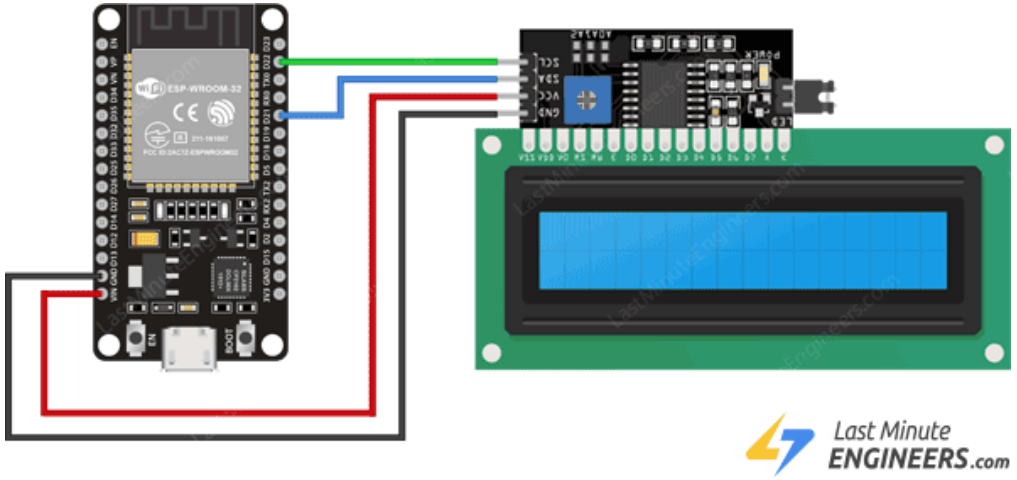


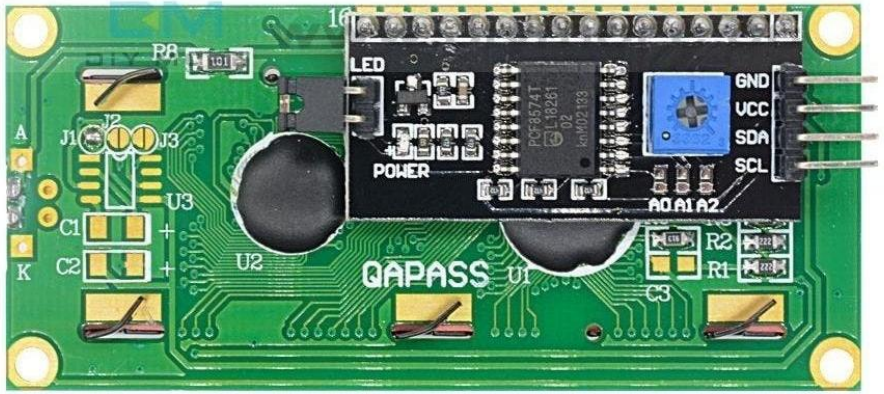


Teachwithict.com

What Can You Do with the ESP32?

- Understand what it can (and cannot) do. Examples:
  - Using an LCD Screen => It can display short messages (but cannot display on monitor via HDMI)







I2C LCD with ESP32 on Arduino IDE - ESP8266 compatible | Random Nerd Tutorials

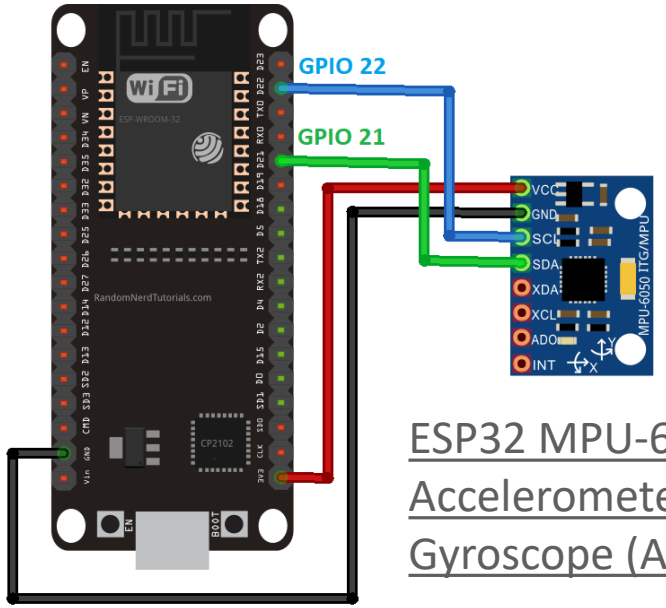
What Can You Do with the ESP32?

- Understand what it can (and cannot) do. Examples:
  - Using an IMU module => It can measure XYZ-axis movement and rotation (but not recognize gestures)

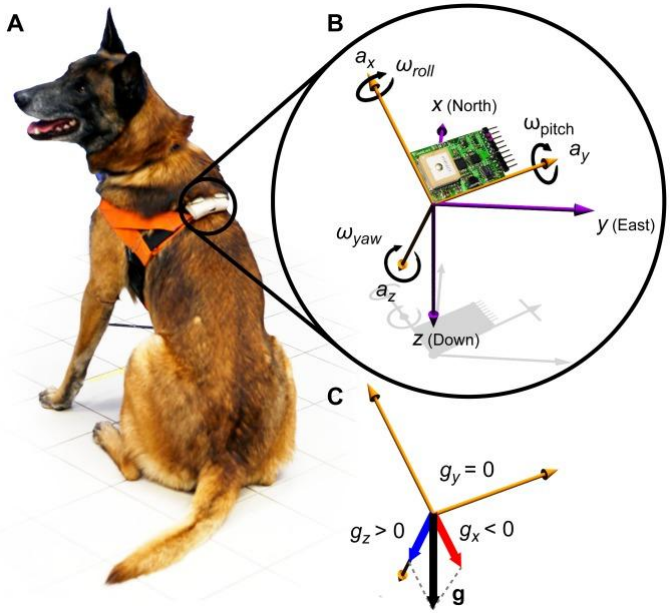








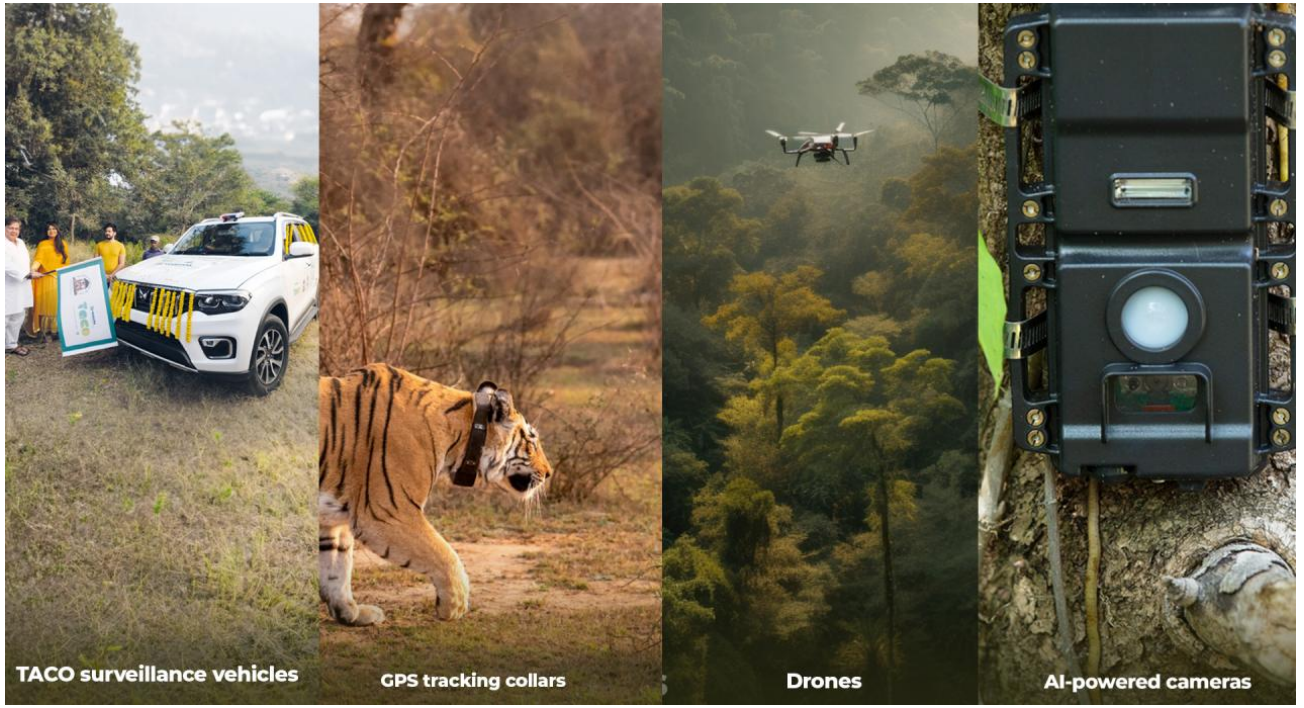
[ESP32 MPU-6050 Accelerometer and Gyroscope (Arduino)]





Action Plan

The goal => Create an IoT Node for defeating Poachers



- What is the form of your Node?
  - Tower, Tree, Animal Wearable, Drone or etc.?

- In this form, what will your Node do?
  - Detect human step (using pressure trap represented by Button)
  - Detect human pass-through (using Ultrasonic Sensor)
  - Detect Gunshot (using Sound Sensor)
  - Detect zero motion (using IMU)
  - Cause Distraction (using LCD)









- How will it connect to the Internet? Only one answer for now - WIFI
  - How to get wireless Internet access in the Jungle?







- How will it communicate with your system in the Cloud? Options:
  - ❖ Blynk – Simplest – One-to-One Dashboard
  - ❖ MQTT – Industry Standard – Only if you did SWE30011 IoT Programming
  - ❖ Direct or REST API calls – different methods depending on the Database or Service
    - Control your Arduino IoT projects with a MongoDB database (donskytech.com)
    - ESP32: Getting Started with Firebase (Realtime Database) | Random Nerd Tutorials

Blynk

mosquitto

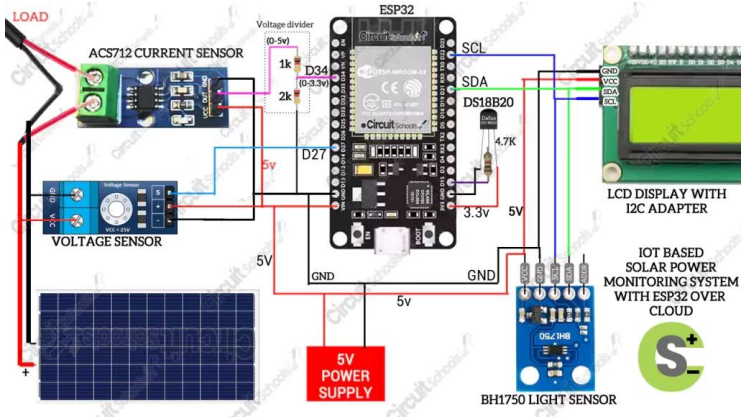
AWS IoT Core

HIVEMQ

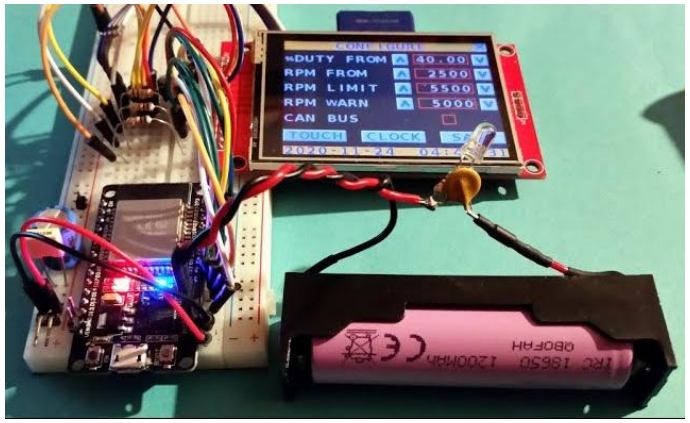
mongoDB.

Firebase

- How will the Node be powered? Only one option for now – USB Powerbank
  - Other options exist!

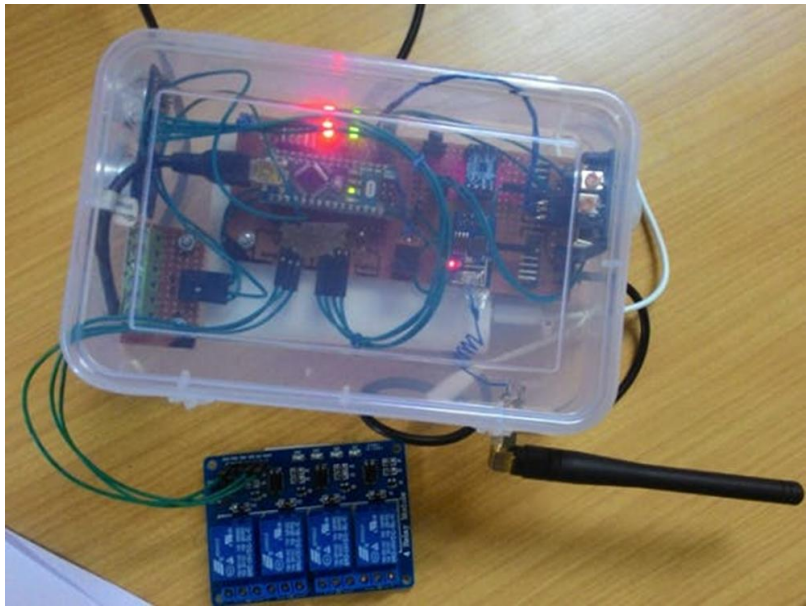








- Finally, how will you secure it? Find a good container to house everything and make it convenient to mount.





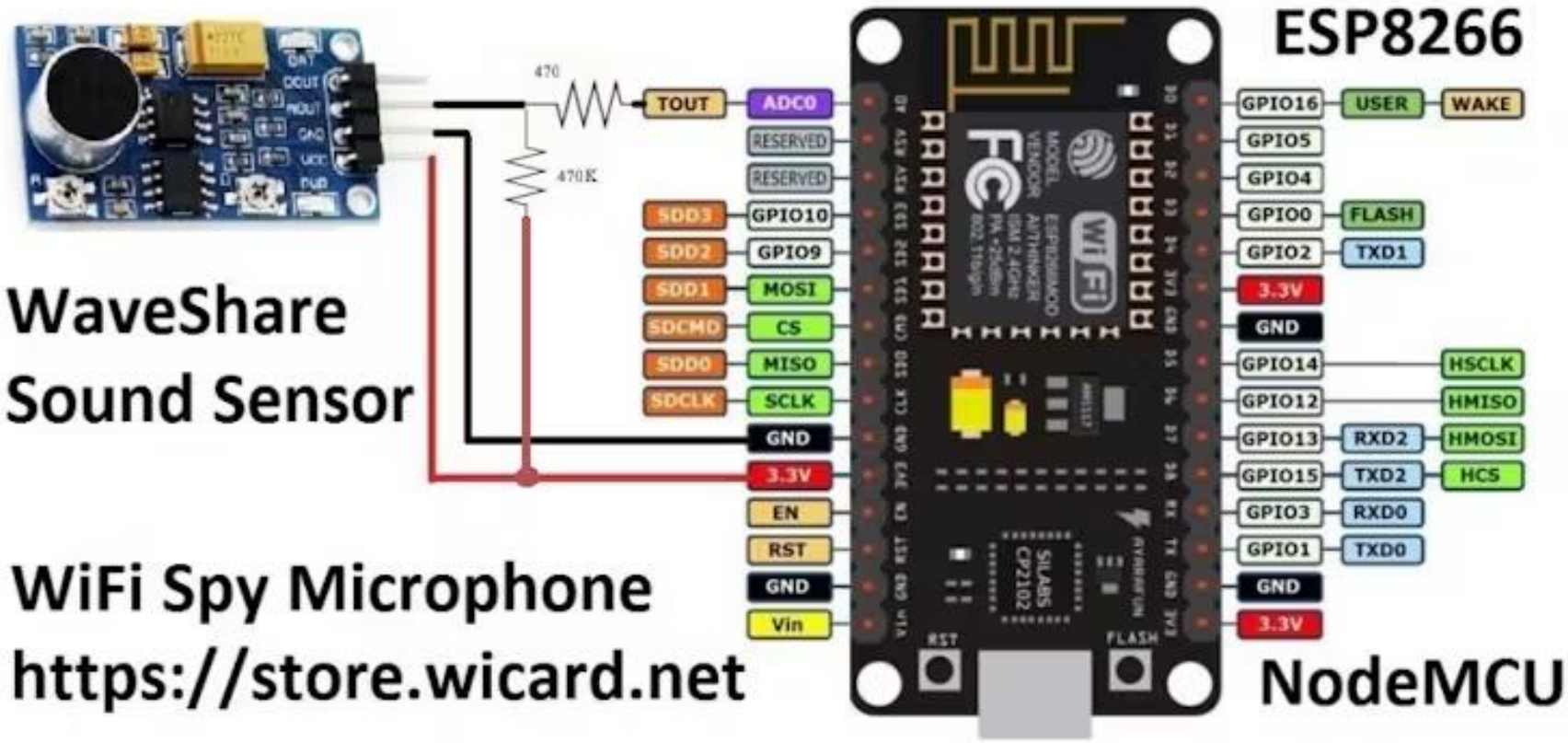


Bonus!

There is a tutorial on using ESP32 + Sound Sensor to record audio and play them from its local web interface.

But beware, the skill requirement is pretty high and you will need to buy your own WaveShare Sound Sensor.

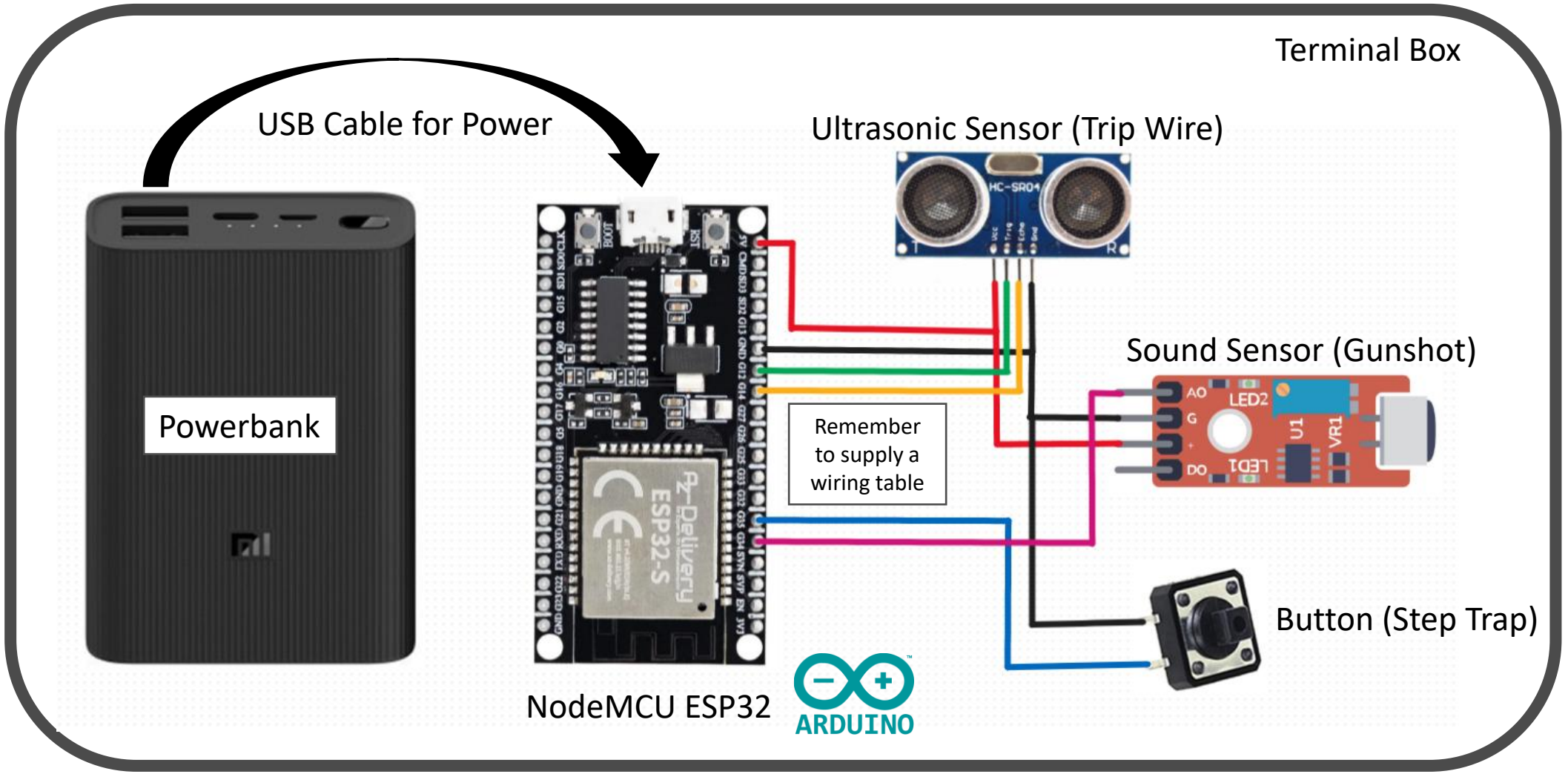
Link: [NodeMCU, ESP8266 and ESP32 Spy Microphone Arduino code - Hackster.io]



Putting It All Together

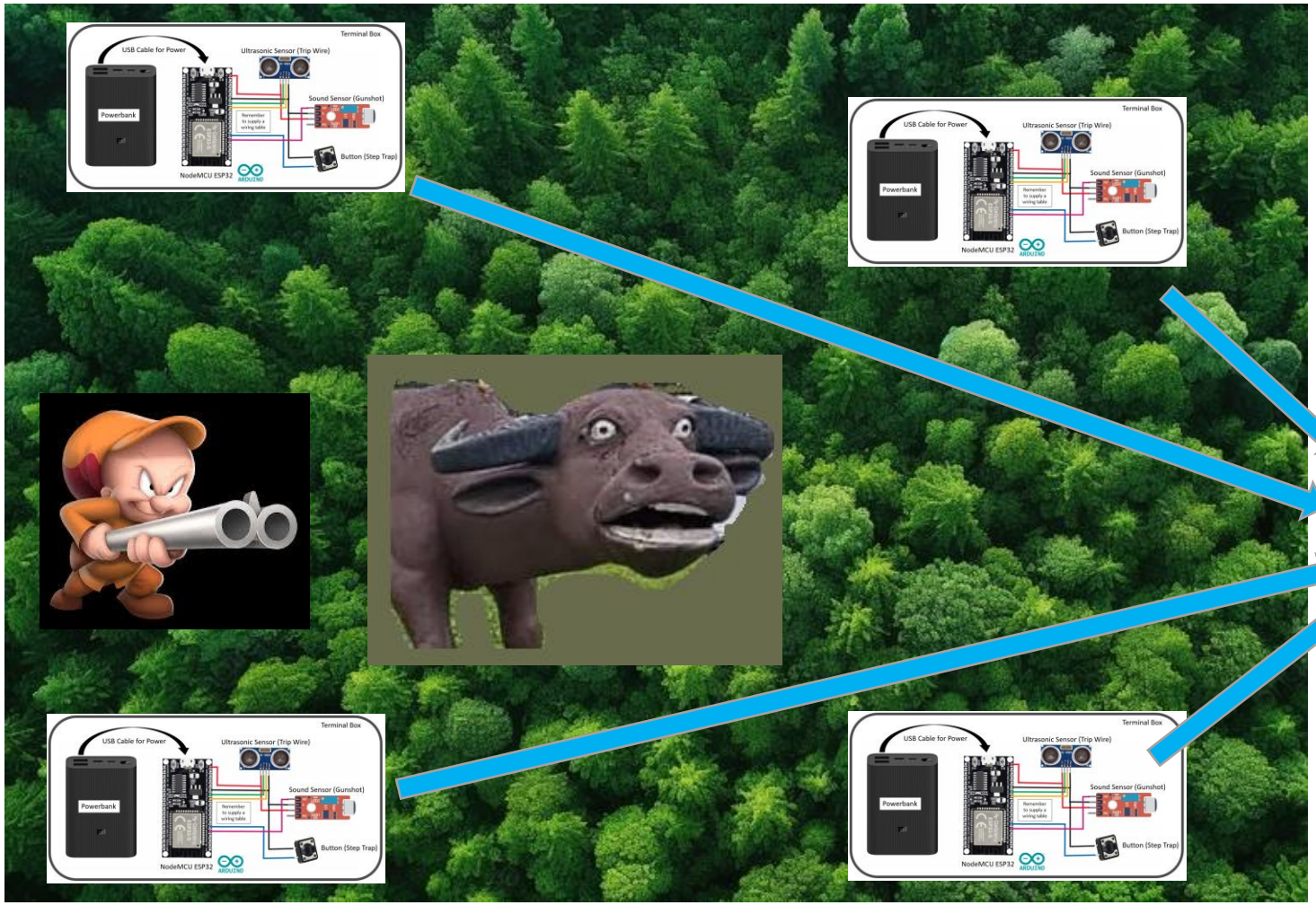
This is an example Block Diagram of my “Poacher Trap” IoT Node Hardware & Software

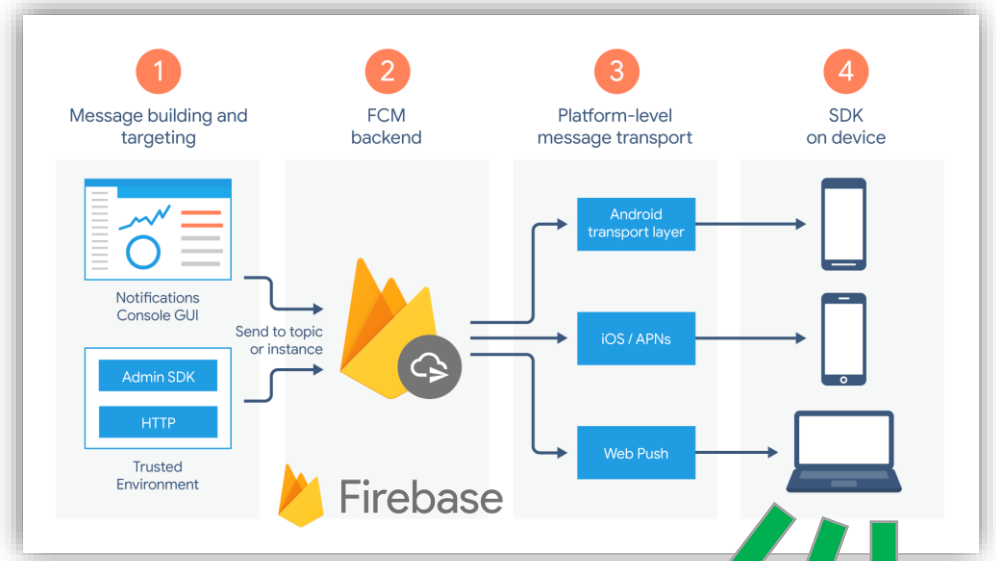
Dr. Mark’s **Poacher Trap Node**



Putting It All Together

This Block Diagram shows how the Nodes are connected to the Internet via WIFI, communicating all readings and alerts to the Rangers via Firebase.





Broadband



Forest-wide WIFI



SWINBURNE UNIVERSITY OF TECHNOLOGY

Consultation for IoT

Every Wednesday @ E404 from 8am-12pm (Except for Tuition Week)

Safety Precaution: Must wear Covered Shoes and Long Pants

Come see me for:

- Technical advice on building your IoT Node.
- Borrow components (if the lab has them) – Sound Sensor, IMU, Temp & Humidity Sensor, Etc.
- Keep the **components separate in their original boxes** when returning.

DON'T FORGET to RETURN ALL HARDWARE in Week 13.



Presented by Dr. Mark Tee Kit Tsun (SUTS)

If you have questions, find me on MS Teams: mtktsun@swinburne.edu.my